

Joseph Robert Livesey

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Research Interests

Solar system dynamics; planetary atmospheres, oceans, interiors, formation, and habitability; open-source software for astrophysics.

Education

University of Wisconsin	Madison, WI
Ph.D. Astronomy	<i>Expected</i>
M.S. Astronomy	<i>Expected</i>
University of Washington	Seattle, WA
B.S. Physics and Astronomy <i>cum laude</i> , with minors in Mathematics and Linguistics	June 2022

Professional Appointments

Graduate Research Assistant, U. Wisconsin	2023–
Research Analyst, U. Washington	2023

Honors and Awards

Fluno Graduate Fellowship, U. Wisconsin	2023
President’s Fellowship, Georgia Institute of Technology (<i>declined</i>)	2023
Phi Beta Kappa	2022
Departmental Honors in Astronomy, U. Washington	2022
Departmental Honors in Physics, U. Washington	2022
Mary Gates Research Scholarship, U. Washington	2020
International Baccalaureate Diploma, Edmonds Woodway High School	2018

Teaching Experience

Teaching Assistant, U. Washington	
The Planets (ASTR 150)	Spring 2023
Reader/Grader, U. Washington	
Computational Methods (ASTR 427)	Autumn 2022
The Planets (ASTR 150)	Winter 2023
Life in the Universe (ASTR/ASTBIO 115)	Spring 2022
Astronomy (ASTR 101)	Winter 2022, Autumn 2022

Service to the Profession

Executive Secretary, NASA XRP	1 panel
SOC, VPlanet Workshop	Sep 2022

Outreach Activities

Co-organizer, Astronomy on Tap Madison	2023–
Volunteer, Washburn Observatory Public Observing	2023–
Co-organizer, AAS 244 Astronomy on Tap	Jun 2024

Memberships

American Astronomical Society (AAS)
AAS Division of Dynamical Astronomy (DDA)
AAS Division of Planetary Sciences (DPS)

Refereed Publications

$h = 1$, first-author: 1, co-author: 0, total citations: 1.

First-Author

2. “Secular Perturbations From Exterior Giants Strongly Influence Gap Complexity in Peas-in-a-Pod Exoplanetary Systems.” [[code](#)]
J. R. Livesey & J. Becker 2024, submitted to AAS Journals.
1. “Orbital Stability and Secular Dynamics of the Proxima Centauri Planetary System.” [[paper](#) | [code](#)]
J. R. Livesey, R. Barnes & R. Deitrick 2024, *ApJ*, 964, 4.

Co-Author

2. “The Fate of Oceans on First-Generation Planets Orbiting White Dwarfs.”
J. Becker, A. Vanderburg & **J. R. Livesey** 2024, submitted to AAS Journals.
1. “History and Habitability of the LP 890-9 Planetary System.” [[code](#)]
R. Barnes et al. **including J. R. Livesey** 2024, submitted to AAS Journals.

Presentations

Contributed talks

3. “Quantifying the Surface Heating of Tidally Migrating White Dwarf Planets.” [[abstract](#)]
AAS 244, Madison, WI. June 2024.
2. “Orbital Dynamics of the Proxima Centauri System.” [[abstract](#)]
AAS 241, Seattle, WA. January 2023.
1. “Orbital and Rotational Evolution of Proxima Centauri b.” [[abstract](#)]
AAS 237, virtual. January 2021.

Posters

1. “Quantifying the Surface Heating of Tidally Migrating White Dwarf Planets.”
Astrobiology Science Conference 2024, Providence, RI. May 2024.

Professional Development

Exoplanets: Compositions, Mineralogy, Evolution (workshop), Chicago, IL

Aug 2024

Technical Skills

Programming: Python, C, Julia

Software: Unix, Git, L^AT_EX, Jupyter, Mathematica, Slurm, MySQL, HTML, CSS

Graduate Coursework

Astronomy: Observational Techniques, Radiative Processes, Astrophysical Fluids,
Stellar Interiors & Evolution, Interstellar Medium, Galaxies, Cosmology,
Solar System Dynamics

Other: Atmospheric Physics, Physical Oceanography, Large-Scale Ocean–Atmosphere Coupling,
Prebiotic Chemistry & Astrobiology

References

Prof. Juliette Becker (U. Wisconsin)

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Prof. Rory Barnes (U. Washington)

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Prof. Zoe Todd (U. Wisconsin)

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Prof. Eric Agol (U. Washington)

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